Supplemental Material for "Information Loss is Local"

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KODAMA VECTOR IN VAIDYA SPACETIME

The ingoing Vaidya metric in advanced Eddington–Finkelstein coordinates (v, r, θ, ϕ) is

$$ds^2 = -f\, dv^2 + 2\, dv\, dr + r^2\, d\Omega^2\,, \tag{1}$$

where $f(v,r) \equiv 1 - 2M(v)/r$ and $M(v)$ is the Bondi mass. The 2D orbit-space metric on the (v,r) submanifold is

$$h_{AB}\, dx^A dx^B = -f\, dv^2 + 2\, dv\, dr\,, \tag{2}$$

with $\det(h) = -1$.

The Kodama vector [1] is defined as $K_A = \epsilon_A{}^B\, \partial_B R$, where $R = r$ is the areal radius and ϵ_{AB} is the volume form on the orbit space: $\epsilon_{vr} = \sqrt{-\det h} = 1$.

Inverse metric.—From $h_{AB} = \left(\begin{smallmatrix} -f & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the inverse is

$$h^{AB} = \begin{pmatrix} 0 & 1 \\ 1 & f \end{pmatrix}. \tag{3}$$

Mixed Levi-Civita tensor.—Raising the second index:

$$\epsilon_v{}^v = \epsilon_{vA}\, h^{Av} = \epsilon_{vr}\, h^{rv} = 1\,, \tag{4}$$
$$\epsilon_v{}^r = \epsilon_{vA}\, h^{Ar} = \epsilon_{vr}\, h^{rr} = f\,, \tag{5}$$
$$\epsilon_r{}^v = \epsilon_{rA}\, h^{Av} = \epsilon_{rv}\, h^{vv} = 0\,, \tag{6}$$
$$\epsilon_r{}^r = \epsilon_{rA}\, h^{Ar} = \epsilon_{rv}\, h^{vr} = -1\,. \tag{7}$$

Covariant components.—Since $\partial_B R = \delta^r_B$:

$$K_v = \epsilon_v{}^B\, \partial_B R = \epsilon_v{}^r = f = 1 - \frac{2M(v)}{r}\,, \tag{8}$$
$$K_r = \epsilon_r{}^B\, \partial_B R = \epsilon_r{}^r = -1\,. \tag{9}$$

Contravariant components and norm.—Raising with h^{AB}:

$$K^v = h^{vA} K_A = K_r = -1\,, \qquad K^r = h^{rA} K_A = K_v + f\, K_r = 0\,. \tag{10}$$

The norm is

$$|K|^2 = h_{AB}\, K^A K^B = h_{vv}\, (K^v)^2 = -f\,, \tag{11}$$

so

$$\boxed{|K| = \sqrt{1 - \frac{2M(v)}{r}}} \quad \text{for } r > 2M(v)\,. \tag{12}$$

The Kodama vector is timelike ($|K|^2 < 0$) for $r > 2M(v)$, null at $r = 2M(v)$, and spacelike for $r < 2M(v)$. Crucially, $|K|$ depends only on the *current* mass $M(v)$ and radius r—it is a local quantity.

The temporal asymmetry parameter is therefore

$$\tau_K(r,v) = 1 - |K| = 1 - \sqrt{1 - \frac{2M(v)}{r}}\,, \tag{13}$$

reaching $\tau_K = 1$ at $r = 2M(v)$, the *apparent horizon.*

EVENT HORIZON IN VAIDYA SPACETIME

The event horizon (EH) is the outermost outgoing null geodesic that just fails to escape to $r \to \infty$. Outgoing radial null geodesics in the Vaidya metric (1) satisfy

$$\frac{dr}{dv} = \frac{1}{2}\left(1 - \frac{2M(v)}{r}\right). \tag{14}$$

The EH is the solution $r_{\rm EH}(v)$ of (14) with boundary condition $r_{\rm EH}(v) \to 2M_{\rm final}$ as $v \to \infty$. This must be integrated *backward* from $v = \infty$—a manifestly teleological procedure.

Step-function collapse

For $M(v) = M_0\, \theta(v)$:

- $v > 0$ (Schwarzschild): $dr/dv = \frac{1}{2}(1 - 2M_0/r)$, static EH at $r = 2M_0$.
- $v < 0$ (Minkowski): $dr/dv = 1/2$, so $r = v/2 + C$. Matching at $v = 0$ gives $r_{\rm EH}(v) = v/2 + 2M_0$.

The EH originates at $r = 0$ when $v = -4M_0$—in flat space, *before* the shell arrives. The apparent horizon appears only at $v = 0$ at $r = 2M_0$.

Self-similar accretion: $M(v) = \lambda v$

The null geodesic equation (14) admits a self-similar ansatz $r(v) = C\, v$:

$$C = \frac{1}{2}\left(1 - \frac{2\lambda}{C}\right) \quad \Longrightarrow \quad 2C^2 - C + 2\lambda = 0\,. \tag{15}$$

For $\lambda < 1/16$, two real roots:

$$C_\pm = \frac{1 \pm \sqrt{1 - 16\lambda}}{4}\,. \tag{16}$$

The event horizon is $r_{\rm EH}(v) = C_- \, v$. The apparent horizon is $r_{\rm AH}(v) = 2\lambda \, v$.

Proof that $C_- > 2\lambda$ (i.e., $r_{\rm EH} > r_{\rm AH}$).—Define $g(\lambda) \equiv C_- - 2\lambda = \frac{1}{4}(1 - \sqrt{1-16\lambda}) - 2\lambda$. At $\lambda = 0$: $g(0) = 0$. The derivative

$$g'(\lambda) = \frac{2}{\sqrt{1-16\lambda}} - 2 = 2\left(\frac{1}{\sqrt{1-16\lambda}} - 1\right) > 0 \quad (17)$$

for all $\lambda > 0$. Since $g(0) = 0$ and $g' > 0$, we have $g(\lambda) > 0$ for $\lambda \in (0, 1/16)$. □

For small λ:

$$\frac{r_{\rm EH}}{r_{\rm AH}} = \frac{C_-}{2\lambda} = 1 + 4\lambda + O(\lambda^2)\,, \quad (18)$$

so the fractional excess grows linearly with accretion rate.

τ_K at the event horizon

At $r = r_{\rm EH} = C_- \, v$ with $M(v) = \lambda v$:

$$\frac{2M(v)}{r_{\rm EH}} = \frac{2\lambda}{C_-}\,. \quad (19)$$

This is a constant (independent of v!). The τ deficit at the EH is

$$\tau_K(r_{\rm EH}) = 1 - \sqrt{1 - \frac{2\lambda}{C_-}} < 1\,. \quad (20)$$

Table I of the main text is computed from Eqs. (16)–(20). For example, at $\lambda = 0.03$: $C_- = (1 - \sqrt{0.52})/4 = 0.0697$, $2\lambda/C_- = 0.861$, $\tau_K = 1 - \sqrt{0.139} = 0.627$.

Smooth accretion: $M(v) = M_0(1 - e^{-v/v_0})$

At late times ($v \gg v_0$), linearization of the null ODE around $r = 2M_0$ gives the particular solution

$$r_{\rm EH}(v) \approx 2M_0 - \frac{2M_0 v_0}{4M_0 + v_0}\, e^{-v/v_0}\,. \quad (21)$$

The difference

$$\Delta r \equiv r_{\rm EH} - r_{\rm AH} = \frac{8M_0^2}{4M_0 + v_0}\, e^{-v/v_0} > 0 \quad (22)$$

confirms $r_{\rm EH} > r_{\rm AH}$ during accretion, with the gap proportional to the accretion rate $dM/dv = (M_0/v_0)\, e^{-v/v_0}$.

EVAPORATING BLACK HOLE

For an evaporating black hole with linearly decreasing mass $M(v) = M_0(1 - v/v_{\rm evap})$, the situation reverses [4]:

$$r_{\rm AH}(v) = 2M(v) > r_{\rm EH}(v)\,. \quad (23)$$

The EH is *inside* the AH because it "knows" the mass will decrease. The Kodama vector is spacelike at $r_{\rm EH}$ (since $r_{\rm EH} < 2M(v)$, the argument of the square root in Eq. (12) is negative), confirming that the EH is inside the trapped region.

The sign of $\Delta r = r_{\rm EH} - r_{\rm AH}$ encodes the direction of mass change:

Scenario	dM/dv	Δr
Accretion	> 0	> 0 (EH outside AH)
Static	$= 0$	$= 0$ (EH = AH)
Evaporation	< 0	< 0 (EH inside AH)

In all cases, $\tau_K = 1$ at the AH, never at the EH. The event horizon requires future knowledge; the apparent horizon (and τ) does not.

CONNECTION TO QUANTUM EXTREMAL SURFACES

The quantum extremal surface (QES) [5, 6] extremizes the generalized entropy

$$S_{\rm gen} = \frac{A}{4G\hbar} + S_{\rm bulk} \quad (24)$$

on a time-slice. In the classical limit ($\hbar \to 0$), $S_{\rm bulk} \to 0$ and the QES reduces to the classical extremal surface, which for spherically symmetric spacetimes is the apparent horizon (the outermost MOTS where $\partial_r A = 0 \Leftrightarrow \theta_+ = 0$).

Including quantum corrections:

$$r_{\rm QES} = r_{\rm AH} + O\left(\frac{\hbar}{M^2}\right), \quad (25)$$

where the correction comes from $\partial_r S_{\rm bulk}$. During evaporation, $\partial_r S_{\rm bulk} > 0$ (bulk entropy increases outward due to Hawking pairs), pushing the QES slightly inward: $r_{\rm EH} < r_{\rm QES} \lesssim r_{\rm AH}$.

The hierarchy is:

$$r_{\rm EH} \;<\; r_{\rm QES} \;\lesssim\; r_{\rm AH} \qquad \text{(during evaporation)}. \quad (26)$$

In the τ framework:

- $\tau_K = 1$ at $r_{\rm AH}$ (classical);
- $\tau_K^{\rm quantum} = 1$ at $r_{\rm QES}$ (quantum-corrected);
- $\tau_K \neq 1$ at $r_{\rm EH}$ in either case.

The τ framework and the QES prescription agree: the operationally relevant boundary for information dynamics is the (quantum-corrected) apparent horizon, not the event horizon.

ECHO TIME DELAY

The tortoise coordinate for the exponential metric $g_{00} = -e^{-r_s/r}$, $g_{rr} = e^{+r_s/r}$ is

$$r^* = \int \frac{\sqrt{g_{rr}}}{\sqrt{-g_{00}}}\, dr = \int e^{r_s/r}\, dr\,. \tag{27}$$

Since $e^{r_s/r}$ is bounded for $r > 0$ (it equals e at $r = r_s$ and approaches 1 as $r \to \infty$), the tortoise coordinate remains finite. This contrasts with Schwarzschild, where $r^* \to -\infty$ as $r \to r_s$.

The round-trip light travel time from the photon sphere ($r = r_s$ in isotropic coordinates) through the wormhole throat and back is

$$\Delta t_{\rm echo} = 2\int_0^{r_s} e^{r_s/r}\, dr \approx 4.2\,\frac{r_s}{c}\,, \tag{28}$$

where the numerical coefficient is evaluated by substituting $u = r_s/r$: $\int_0^{r_s} e^{r_s/r}\, dr = r_s \int_1^\infty u^{-2}\, e^u\, du$. The integral converges and equals approximately $2.1\, r_s$, giving a round trip of $\approx 4.2\, r_s/c$.

For a $60\, M_\odot$ remnant: $r_s = 2GM/c^2 = 177$ km, $\Delta t \approx 4.2 \times 177/c \approx 2.5$ ms, at a frequency ~ 400 Hz—squarely in the LIGO band.

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